NURSESYNC

Product Requirements Document: Agent Build

Agent name: NurseSync, a team of 4 AI agents for hospital nurse scheduling

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1. PROBLEM

Nurse managers at hospitals spend hours every week building schedules by hand, because staffing needs change daily and nurse availability, time-off requests and staffing rules live in different places. The result is understaffed shifts, extra overtime, rule violations that are caught late, and tired nurses.

Supporting Context (optional)

- A manager must balance patient demand, each nurse's availability and approved time off, and rules like rest between shifts, overtime limits and the right skills on each unit, all at the same time.
- Call-outs and swap requests arrive after the schedule is posted, so the manager has to rebuild part of it quickly.
- Schedules built by hand are hard to audit, so rule violations are often found only after the schedule is posted.

1a. Opportunity

Give every nurse manager a draft schedule that already follows the rules, so they review and approve it instead of building it. This returns hours each week to patient care and cuts down on last-minute gaps and overtime.

Size of the Opportunity

- A single weekly schedule for one hospital unit can take a manager hours to build and check by hand.
- Every hour saved per manager each week goes back to patient care and staff support, and fewer rule violations means less overtime cost and less risk.

1b. Users & Needs

Primary user(s): Nurse managers and charge nurses who build the weekly schedule for one hospital unit. They care most about full coverage, fairness and no rule violations.

Secondary users: Staff nurses, who enter availability and request time off, and hospital administrators, who care about overtime cost and compliance.

Key User Needs

As a nurse manager, I need a draft schedule that already follows the rules because checking every rule by hand takes too long.

As a nurse manager, I need to see why the schedule was built this way because I have to trust it before I approve it.

As a staff nurse, I need my availability and approved time off respected because I plan my life around my shifts.

As a staff nurse, I need swap and call-out changes handled quickly and fairly because I need predictable time away from work.

2. PROPOSED SOLUTION

NurseSync is a team of four AI agents for hospital nurse scheduling. It runs every week, and again whenever a nurse calls out. A Demand agent estimates how many nurses each day needs, an Availability agent collects who can work and who asked for time off, a Scheduler agent builds a draft, and a Rules Checker agent reviews the draft and sends it back until it follows the rules. The nurse manager then reviews and approves the schedule. The agents inform the human, and the human makes the call.

2a. Value Proposition

Nurse managers who struggle to build compliant schedules, because demand, availability and staffing rules are scattered, use NurseSync, a team of AI agents that drafts and checks the weekly schedule. Unlike building it by hand in a spreadsheet, it checks every rule before the manager sees the draft, helping them approve a schedule in minutes instead of building one for hours.

2b. Top 3 MVP Value Props

The Vitamin *(must-have baseline)*: Every shift in the week is covered by an available, qualified nurse, with no one scheduled outside their availability or approved time off.

The Painkiller *(solves the core pain)*: No more building the schedule by hand. The manager gets a full draft that has already passed every rule check.

The Steroid *(the magic moment)*: When a nurse calls out, NurseSync suggests replacements in minutes, with the reason for each, ready for the manager to approve.

2c. Success Metrics

Goal	Signal	Metric	Target

Save manager time	Managers stop building schedules by hand	Minutes from request to approved weekly schedule	Under 30 minutes
Full coverage	Every shift has the nurses it needs	Percent of shifts fully staffed in the approved schedule	At least 95%
Trustworthy drafts (quality)	Drafts follow every rule	Rule violations in drafts shown to the manager	0

3. AGENT REQUIREMENTS

3a. Tools

Tool name	What it does	API it calls	Data it returns
get_demand	Estimates how many nurses each unit needs per shift (Demand agent)	Shift Board data: sample patient counts per unit	Unit, date, shift, nurses needed
get_availability	Reads who can work when and who asked for time off (Availability agent)	Shift Board data: availability and leave	Nurse, available times, approved time off
get_roster	Reads each nurse's unit, skills and hours worked (Scheduler agent)	Shift Board data: roster	Name, unit, certifications, hours this week
check_rules	Tests a draft against rest-time, overtime and skill-mix rules (Rules Checker agent)	Rules table in our own code	Pass or fail, and each violation found

3b. System Prompt v0

You are NurseSync, a scheduling assistant for a hospital nurse manager. Each week, or when a nurse calls out, work through these steps.

1. Call get_demand to find how many nurses each unit needs per shift, and get_availability to find who can work and who has approved time off. Do these at the same time.

2. Call get_roster for each nurse's unit, skills and hours worked this week. Build a draft schedule that fills every shift using only available, qualified nurses.

3. Call check_rules on the draft. If it fails, fix each violation and check again. Repeat up to 3 times.

4. Give the manager the passing draft, a list of any open gaps, and one line of reasoning for each hard choice.

Constraints: Never schedule a nurse outside their availability or approved time off. Never invent data. Never send anything to nurses; the manager approves and decides what happens next. If data is missing for a nurse or a day, list it under "Needs manual review" instead of guessing.

Escalation: If the draft still fails the rules after 3 tries, stop and hand it to the manager with the open violations.

Output format: a weekly schedule by day and shift, then open gaps, then the reasoning notes.

3c. Blast Radius

Worst-case scenario: NurseSync produces a schedule that breaks a rule or leaves a shift uncovered, and the manager approves it without noticing. Nurses on the unit are affected, but the damage can be undone: the schedule can be corrected and reissued. The blast radius stays small because every tool is read-only, nothing is sent to nurses without the manager, and the project only uses fake data.

Failure Modes & Safeguards

Failure mode	Worst-case impact	Safeguard
Scheduler books a nurse who is unavailable or on approved leave	A shift is uncovered and a nurse is upset	Rules Checker tests availability; manager approves before anything goes out
Draft breaks rest-time or overtime rules	Safety and compliance risk	Rules Checker loop; stop after 3 tries and hand the open violations to the manager
Demand estimate is too low	A unit is understaffed	Manager sees the demand numbers and can change them before approving
Data is missing for a nurse or a day	A nurse or shift is left out	Listed under "Needs manual review"; the agents never guess

3d. Eval Card

Case	Input	Expected output — written before you run

1 — Golden example (normal input)	12 nurses on one unit, one week, clear availability, no conflicts	Every shift is covered, 0 rule violations, no one scheduled outside availability, and a short reasoning note on any hard choice
2 — Golden example (edge case)	Two nurses on approved leave and one called out mid-week, leaving fewer nurses than the demand needs	Draft covers as many shifts as possible, lists the exact open gaps, and suggests replacement options for the manager to approve
3 — Adversarial input	Availability data is empty for one nurse, and get_demand returns an error for one day	The nurse and the day appear under "Needs manual review"; no invented data, no crash, and a clear message to the manager